

Crossmark

PAPER

RECEIVED
dd Month yyyy
REVISED
dd Month yyyy

Hierarchical Tactical Corridor Planning and Control for Wheel-to-Wheel Autonomous Racing

Shuyi Gao, Defu Lin and Ren Jin

School of Aerospace Engineering, Beijing Institute of Technology, Beijing, China

Keywords: Autonomous racing, model predictive contouring control, second-order cone programming, reinforcement learning, Frenet frame, friction circle, game-theoretic motion planning.

Abstract

Wheel-to-wheel autonomous racing couples strategic interaction, real-time trajectory optimization, and friction-limit control. Solving these decisions in a single monolithic game or nonlinear program is computationally fragile at Formula-class speeds, whereas end-to-end neural control is difficult to audit on a safety-critical vehicle. This paper presents a hierarchical architecture that restricts learning to tactical corridor synthesis and retains optimization-based planning and control downstream. A map-conditioned Truncated Quantile Critics policy and a hysteretic finite-state machine select an overtake side, velocity cap, and time-varying Frenet corridor. A sparse second-order cone program computes a curvature-regularized lateral path within that corridor, a forward-backward sweep imposes a super-elliptic GG friction envelope, and a nonlinear model predictive contouring controller tracks the resulting reference with a terminal progress reward and a soft rear-axle friction constraint. Closed-loop simulations on the Dallara EAV24 at Yas Marina demonstrate two successful fast-closure overtakes, with pass times of 1.13 s and 8.63 s after entering OVERTAKE, peak speed up to 73.10 m/s, and lateral RMS tracking errors of 0.02 and 0.05 m. The results support the central premise that learning should shape the feasible set, while model-based layers enforce vehicle dynamics and handling-limit execution.

1 Introduction

Autonomous racing at the vehicle handling limit is among the most demanding closed-loop control problems in robotics. A Formula-class car must decide *whether* to overtake, *which side* to attempt, *where* to place the resulting trajectory within the track boundaries, and *how* to execute it at the friction limit—all while an adversarial opponent simultaneously optimizes its own plan. These four decisions span discrete game theory, continuous trajectory optimization, and high-bandwidth nonlinear control, operating across timescales that differ by three orders of magnitude. The Abu Dhabi Autonomous Racing League (A2RL), with its 5.28 km Yas Marina Formula 1 circuit and the 700 kg Dallara EAV24 platform, provides a uniquely demanding testbed for this hierarchical challenge [9–11].

Classical autonomous driving architectures decompose the problem into perception, behavior planning, motion planning, and control [37, 38]. The behavior planner selects a discrete maneuver—follow, change lane, yield—which the motion planner translates into a continuous trajectory [42, 43]. In racing, however, the motion planner has traditionally been cast as a single-vehicle, offline lap-time optimization problem [1–5], producing a fixed raceline tracked by an MPC or MPCC controller [6, 7, 13, 15, 16]. Full-stack racing platforms developed for the Indy Autonomous Challenge and A2RL document this paradigm in detail [8, 10–12]. The fundamental limitation is that a pre-computed raceline cannot adapt to an opponent who is simultaneously optimizing their own trajectory: an adversary that defends the apex creates a geometric deadlock that a single-vehicle planner cannot resolve without a higher-level instruction about which side to attempt the pass and when to commit.

Recognising that wheel-to-wheel racing is inherently a non-cooperative dynamic game, several groups have proposed game-theoretic formulations of the tactical layer. Wang *et al.* [21] cast head-to-head racing as an iterative best-response game with local Nash convergence guarantees. Iterative linear-quadratic game solvers [23, 24] compute approximate generalized feedback Nash equilibria at near-real-time rates. Real-time game-theoretic planners [25] and viability-theoretic approaches [16, 26] provide formal safety certificates, while sampling-based MPPI [27] and Bayesian

intent inference [29, 36] address opponent-model uncertainty. Despite this progress, pure game-theoretic methods face four practical obstacles in Formula-class deployment: (i) exact Nash equilibria of continuous-time differential games are intractable, and approximate solvers [23, 24] may not converge within a single planning cycle at circuit speeds; (ii) the Nash solution is fragile to misspecification of the opponent’s cost function; (iii) the discrete decision to switch from shadowing to overtaking is not naturally captured by smooth equilibrium computations; and (iv) hand-tuned weights and horizon lengths must be recalibrated for each track and pace configuration [39, 40].

In parallel, data-driven methods have emerged as an alternative. Sony’s GT Sophy [28] and subsequent RL-for-racing work [30, 31, 41] demonstrated that end-to-end neural policies can achieve superhuman performance in simulation. Model-based RL [22, 33], safe RL [32, 34, 45], and learning-from-demonstration [44, 47] have further expanded the data-driven toolkit. Lin *et al.* [46] compared optimal control and RL at the handling limit, finding that RL excels in multi-agent settings but underperforms in single-vehicle lap time. However, deploying end-to-end neural controllers on a physical Formula car raises four serious concerns: (i) the EAV24’s nonlinear tire model, aerodynamic downforce map, and differential torque vectoring are not faithfully reproduced in open simulators; (ii) a neural network that outputs steering torque cannot be audited or certified by race engineers; (iii) actor-critic policies produce high-frequency command jitter that degrades tire wear; and (iv) without an explicit spatial representation, the network must re-discover track geometry from reward signals alone [13, 47].

We resolve these tensions through architectural separation: the RL policy is restricted to the tactical layer, leaving model-based solvers for planning and control. A Truncated Quantile Critics (TQC) policy outputs a low-dimensional action vector that reshapes the feasible lateral corridor $\mathcal{C}_k = [L_{\min,k}, L_{\max,k}]$ and the FSM hysteresis thresholds, without ever modifying the planner cost or the controller dynamics. Critically, the TQC observation vector embeds a *fixed-track prior*: a frozen, map-level encoding of raceline curvature, track banking, and friction-limit speed at each arc-length station. This spatial prior supplies the policy with geometric context, so the learned tactical decision is conditioned on upcoming curvature and available track width rather than on reward history alone. A heuristic FSM translates the continuous TQC output into a discrete mode sequence and a corridor-carving decision, delivering a constraint triple $(\mathcal{C}_k, v_{\text{cap}}, \text{mode})$ to the planner.

Given the corridor, the planning layer solves a sparse second-order cone program (SOCP) for a curvature-smooth lateral path, followed by a forward–backward speed sweep on a super-elliptic GG diagram with empirically identified exponent $p = 0.75$. This two-stage decomposition is substantially faster than monolithic nonlinear trajectory optimization [5] while naturally capturing the coupled longitudinal–lateral grip limits. The resulting reference is tracked by a nonlinear Model Predictive Contouring Controller (MPCC) that augments an 8-state dynamic bicycle model with a path-progress state; a terminal progress reward automatically generates corner entry-braking and exit-acceleration without per-corner reference shaping, and a soft rear-axle friction-cone constraint prevents tire saturation without rendering the nonlinear program infeasible. All three layers share a common Frenet representation.

The contributions of this paper are:

1. **Map-conditioned tactical corridor synthesis.** A TQC policy receives a frozen preview of curvature, track width, and friction-limited speed and outputs only interpretable tactical quantities: side preference, safety margin, velocity cap, and corridor bounds. A hysteretic FSM converts this output into a time-varying feasible set $\mathcal{C}_k$, preventing the learned policy from directly commanding actuators or altering downstream costs.
2. **Convex local path planning with GG-constrained speed profiling.** A sparse SOCP over Frenet state and curvature slack produces a curvature-regularized path inside the tactical corridor. A forward–backward sweep on a super-elliptic GG envelope then constructs the corresponding grip-limited speed profile, separating geometric feasibility from speed feasibility while preserving their coupling through curvature.
3. **Closed-loop validation of the hierarchy in head-to-head racing.** An 8-state dynamic-bicycle MPCC with a terminal progress reward and a soft rear-axle friction constraint executes the planned references. Two closed-loop Yas Marina fast-closure overtake simulations quantify pass time, tactical gain, peak speed, and tracking error, showing that the same planner and controller can execute distinct tactical corridors without case-specific retuning.

The remainder of the paper is organized as follows. Section 2 derives the unified vehicle and

Frenet models. Section 3 presents the RL-driven tactical layer. Section 4 details the SOCP path planner and the FB speed profiler. Section 5 formulates the MPCC controller. Section 6 discusses inter-layer signaling. Section 7 reports two closed-loop overtake simulations. Section 8 concludes.

2 Vehicle and Frenet Modeling

We adopt the convention that the controller sees a body-frame dynamic single-track model (used inside MPCC), while the planner sees a Frenet kinematic model derived from the same physical vehicle. Both reduce to a common parameterization in the limit of zero side-slip.

2.1 Body-Frame Dynamic Single-Track Model

Let $\mathbf{x} \in \mathbb{R}^8$ and $\mathbf{u} \in \mathbb{R}^2$ denote the MPCC state and input,

$$\mathbf{x} = [p_x, p_y, \psi, v_x, v_y, r, \delta, s]^\top, \quad \mathbf{u} = [a, \dot{\delta}]^\top, \quad (1)$$

where (p_x, p_y) is the position in the inertial frame attached to the *prediction window origin*, ψ is the yaw angle, (v_x, v_y) are the body-frame longitudinal and lateral velocities, $r = \dot{\psi}$ is the yaw rate, δ is the front steering angle, s is an integrated longitudinal-progress state used by the contouring cost, a is the commanded longitudinal acceleration and $\dot{\delta}$ is the steering rate. The progress state is integrated as $\dot{s} = v_x$, so s acts as the path parameter inside the contouring formulation of Sec. 5.

The kinematic relations are

$$\dot{p}_x = v_x \cos \psi - v_y \sin \psi, \quad \dot{p}_y = v_x \sin \psi + v_y \cos \psi, \quad \dot{\psi} = r. \quad (2)$$

Front and rear slip angles are obtained with the standard small-angle single-track expressions, regularized at low speed by an offset $\varepsilon_v = 0.5 \text{ m/s}$,

$$\alpha_f = \arctan\left(\frac{v_y + l_f r}{v_x + \varepsilon_v}\right) - \delta, \quad (3)$$

$$\alpha_r = \arctan\left(\frac{v_y - l_r r}{v_x + \varepsilon_v}\right). \quad (4)$$

Linear cornering stiffness yields the lateral tire forces

$$F_{yf} = -C_f \alpha_f, \quad F_{yr} = -C_r \alpha_r. \quad (5)$$

Substituting into Newton–Euler gives the dynamic equations

$$\dot{v}_x = a + r v_y - \frac{1}{m} F_{yf} \sin \delta, \quad (6)$$

$$\dot{v}_y = \frac{1}{m} (F_{yf} \cos \delta + F_{yr}) - r v_x, \quad (7)$$

$$\dot{r} = \frac{1}{I_z} (l_f F_{yf} \cos \delta - l_r F_{yr}), \quad (8)$$

$$\dot{\delta} = \dot{\delta}. \quad (9)$$

The full continuous dynamics are written compactly as

$$\dot{\mathbf{x}} = f_c(\mathbf{x}, \mathbf{u}), \quad (10)$$

and discretized with explicit Runge–Kutta of order four with sample $T_s = 0.05 \text{ s}$,

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k) \triangleq \text{RK4}(f_c, \mathbf{x}_k, \mathbf{u}_k, T_s). \quad (11)$$

Vehicle parameters used in the experiments are consolidated in Table 1.

2.2 Frenet Kinematic Model and GG Envelope

The reference centerline (or raceline) is a smooth curve $\mathbf{r}(s) \in \mathbb{R}^2$ parameterized by arc-length $s \in [0, L_{\text{trk}}]$, with curvature $\kappa(s)$ and tangent angle $\theta(s)$ such that $\theta'(s) = \kappa(s)$. Any point in a tube around $\mathbf{r}$ admits the Frenet decomposition

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \mathbf{r}(s) + l \mathbf{n}(s), \quad \mathbf{n}(s) = [-\sin \theta(s), \cos \theta(s)]^\top, \quad (12)$$

where l is the signed lateral offset and $\alpha \triangleq \psi - \theta(s)$ the heading deviation. Differentiating (12) and projecting onto the tangent/normal frame yields, for a nonholonomic body with body speed v ,

$$\dot{s} = \frac{v \cos \alpha}{1 - l \kappa(s)}, \quad (13)$$

$$\dot{l} = v \sin \alpha, \quad (14)$$

$$\dot{\alpha} = \dot{\psi} - \kappa(s) \dot{s}. \quad (15)$$

Spatial reformulation Replacing time by arc-length and using $\frac{d}{ds} = \frac{1}{s} \frac{d}{dt}$,

$$l'(s) = \frac{(1-l\kappa)\tan\alpha}{1}, \quad \alpha'(s) = \frac{(1-l\kappa)\dot{\psi}}{v\cos\alpha} - \kappa(s). \quad (16)$$

Linearizing (16) around a nominal $(l_i, \alpha_i, \kappa_i)$ provides the planner-side dynamics

$$\frac{d}{ds} \begin{bmatrix} l \\ \alpha \end{bmatrix} \approx \underbrace{\begin{bmatrix} -\kappa_i \tan \alpha_i & \frac{1-\kappa_i l_i}{\cos^2 \alpha_i} \\ 0 & 0 \end{bmatrix}}_{\mathbf{A}_i} \begin{bmatrix} l \\ \alpha \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & 0 \\ \frac{1-\kappa_i l_i}{v_{\text{ref}}^2 \cos \alpha_i} & 0 \end{bmatrix}}_{\mathbf{B}_i} \begin{bmatrix} \zeta \\ \sigma \end{bmatrix} + \mathbf{b}_i, \quad (17)$$

which is the building block of the SOCP discretization in Sec. 4. The auxiliary variables (ζ, σ) are the curvature slack and its second-order cone epigraph variable.

GG envelope The combined-slip envelope of the EAV24 is well captured by the super-elliptic relation

$$\left| \frac{a_y}{a_y^{\max}(v)} \right|^p + \left| \frac{a_x}{a_x^{\max}(v)} \right|^p \leq 1, \quad (18)$$

where $a_y^{\max}(v)$ and $a_x^{\max}(v)$ are speed-dependent grip caps obtained from a vehicle data table, and $p = 0.75$ is identified empirically from on-track lateral/longitudinal traces. Solving (18) for the available longitudinal acceleration gives

$$a_x^{\text{avail}}(v, a_y) = a_x^{\max}(v) \left[1 - \left(\frac{|a_y|}{a_y^{\max}(v)} \right)^p \right]^{1/p}, \quad (19)$$

which is the central nonlinearity of the FB speed planner. Aerodynamic drag is modeled as a quadratic function of speed,

$$a_{\text{drag}}(v) = \frac{1}{m} (c_{d,2} v^2 + c_{d,1} v + c_{d,0}), \quad (20)$$

whose coefficients $(c_{d,2}, c_{d,1}, c_{d,0})$ are identified from coastdown data. Drag is added to the longitudinal force balance with the sign convention of (6).

3 Tactical Layer: RL-Driven Corridor Synthesis

The tactical layer transforms the multi-agent racing scene into a spatio-temporally smooth feasible corridor for the planner. It is the only layer that observes opponents, and it communicates with the optimizer through constraints rather than through cost weights or low-level commands. This interface keeps the learned component interpretable: changing the policy changes where the optimizer is allowed to drive, not how the optimizer values tracking, smoothness, or vehicle dynamics.

3.1 Hybrid TQC Policy, Track Prior and Finite-State Machine

Observation At each tactical step the environment supplies a compact dynamic observation $\mathbf{o}_t \in \mathbb{R}^{27}$. The ego-vehicle block comprises normalized speed, heading error, lateral offset, and body-frame accelerations,

$$\mathbf{o}_t^{\text{ego}} = [\bar{V}, \chi, \bar{n}, \bar{a}_x, \bar{a}_y, \bar{s}_e]^\top, \quad (21)$$

where each component is scaled by its characteristic range (e.g. $\bar{V} = V/V_{\text{char}}$, $\bar{n} = n/n_{\text{char}}$); the numerical scaling constants are consolidated in Table 1. The nearest opponent in wrapped arc-length contributes a symmetrically normalized block,

$$\mathbf{o}_t^{\text{opp}} = [\bar{\Delta}s, \bar{\Delta}n, \bar{\Delta}V_s, \bar{s}_o, \bar{n}_o, \bar{g}, \bar{\tau}_{\text{tc}}, \bar{V}_o]^\top, \quad (22)$$

with the wrapped relative progress

$$\Delta s = ((s_o - s_e + L_{\text{trk}}/2) \bmod L_{\text{trk}}) - L_{\text{trk}}/2, \quad \Delta n = n_o - n_e, \quad (23)$$

$\Delta V_s = V_o - V_e$, the Euclidean gap $g = \sqrt{\Delta s^2 + \Delta n^2}$, and the time-to-collision $\tau_{\text{tc}} = \Delta s / \max(-\Delta V_s, \varepsilon_\tau)$ for $\Delta V_s < 0$. Three one-hot closing-speed buckets $\mathbf{b}_t \in \{0, 1\}^3$, the locked overtake-side indicator $\sigma_t^{\text{lock}} \in \{-1, 0, +1\}$, and the FSM-mode one-hot encoding $\mathbf{e}_{m_t} \in \{0, 1\}^{|\mathcal{M}|}$ complete the observation vector

$$\mathbf{o}_t = [\mathbf{o}_t^{\text{ego}}, \mathbf{o}_t^{\text{opp}}, \mathbf{b}_t, \sigma_t^{\text{lock}}, \mathbf{e}_{m_t}] \in \mathbb{R}^{27}. \quad (24)$$

Neural fixed-track prior The policy feature extractor owns a fixed map of the circuit inside the neural network. A non-trainable lookup table $\mathcal{P} = \{\rho_i\}_{i=0}^{M-1}$ is constructed offline from the baseline raceline before training. Querying this table at arc-length s returns a compact encoding of the track geometry over the next d_p metres,

$$\rho(s) = [\bar{\kappa}_0, \bar{\kappa}_{0:d_1}^{\max}, \bar{\kappa}_{d_1:d_2}^{\max}, \bar{\kappa}_{d_2:d_3}^{\max}, \bar{\kappa}_{d_3:d_p}^{\max}, \bar{\kappa}_{\text{mean}}, \bar{\kappa}_{\text{std}}, \bar{W}_{\min}^L, \bar{W}_{\min}^R, \bar{W}_0^L, \bar{W}_0^R, \Delta \bar{W}_{\min}]^\top, \quad (25)$$

where the first five entries are the local curvature and the peak absolute curvatures in successive d_{win} -metre windows; the remaining entries encode current and bottleneck left/right track half-widths, each normalized by the characteristic track width W_{char} . The actor and critics therefore receive the map-conditioned input

$$\tilde{\mathbf{o}}_t = [\mathbf{o}_t, \rho(s_e), \rho(s_o)] \in \mathbb{R}^{d_o+2d_p}, \quad (26)$$

processed by a feedforward feature network. Critically, the track map is embedded in the neural network as a fixed spatial prior indexed by ego and opponent arc-length; it is not a hand-coded planner cost term.

Action The TQC actor outputs a bounded vector $\mathbf{u}_t \in [-1, 1]^{d_a}$, which is affine-mapped elementwise to the physical action

$$\mathbf{a}_t = \underline{\mathbf{a}} + \frac{1}{2}(\mathbf{u}_t + \mathbf{1}) \odot (\bar{\mathbf{a}} - \underline{\mathbf{a}}), \quad (27)$$

where the action vector

$$\mathbf{a}_t = [L_t^{\text{cap}}, R_t^{\text{cap}}, d_{\text{safe}}, V_{\text{cap}}, g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}, b_{\text{lat}}, w_{\text{safe}}, q_{\text{side}}]^\top \quad (28)$$

comprises the left/right corridor caps ($L_t^{\text{cap}}, R_t^{\text{cap}}$), the safety margin d_{safe} , the velocity cap V_{cap} , the FSM hysteresis thresholds ($g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}$), a lateral bias b_{lat} , a corridor-width bonus w_{safe} , and a side-selection score $q_{\text{side}} \in [-1, 1]$ whose sign determines the overtake direction ($q_{\text{side}} \geq 0$: left; $q_{\text{side}} < 0$: right). The side decision is locked for the duration of one duel. The elementwise bounds $\underline{\mathbf{a}}, \bar{\mathbf{a}}$ are reported in Table 1. After affine mapping, consistency clips

$$\begin{aligned} g_{\text{chase}} &\geq g_{\text{ot}} - \varepsilon_{co}, & g_{\text{ot}} &\geq g_{\text{chase}} - \varepsilon_{oc}, \\ g_{\text{abort}} &\geq g_{\text{chase}} + \varepsilon_{ca}, & L_t^{\text{cap}} + R_t^{\text{cap}} &\geq \varepsilon_{\text{corr}}. \end{aligned} \quad (29)$$

are enforced with small positive offsets ($\varepsilon_{co}, \varepsilon_{oc}, \varepsilon_{ca}, \varepsilon_{\text{corr}}$) to maintain a non-empty corridor and a well-ordered FSM throughout training.

Finite-state machine The discrete tactical mode $m_t \in \mathcal{M} = \{\text{RACELINE}, \text{SHADOW}, \text{OVERTAKE}, \text{HOLD}, \text{DEFEND}\}$ is selected by a hysteretic FSM. Let g denote the longitudinal gap to the leading opponent and g_{rear} the gap to the closest trailing opponent. The FSM transition logic is

$$m_{t+1} = \begin{cases} \text{OVERTAKE} & g \leq g_{\text{ot}} \wedge \text{ready} \wedge \bar{c} \\ \text{SHADOW} & g_{\text{ot}} < g \leq g_{\text{chase}} \\ \text{HOLD} & m_t = \text{OVERTAKE} \wedge \text{abort} \\ \text{DEFEND} & g > g_{\text{def}}^{\text{front}} \wedge 0 < g_{\text{rear}} < g_{\text{def}}^{\text{rear}} \wedge \bar{c} \\ \text{RACELINE} & g > g_{\text{chase}}, \end{cases} \quad (30)$$

where $\bar{c}$ indicates that the post-overtake immunity window has expired. The abort criterion is $(g > g_{\text{abort}}) \vee (\Delta n_{\text{min}} < \Delta n_{\text{safe}} \wedge g < g_{\text{interlock}})$, which guards against late-corner divergence and lateral interlock with the opponent. The thresholds ($g_{\text{chase}}, g_{\text{ot}}, g_{\text{abort}}$) are the RL-modulated hysteresis parameters from $\mathbf{a}_t$, while $(g_{\text{def}}^{\text{front}}, g_{\text{def}}^{\text{rear}}, \Delta n_{\text{safe}}, g_{\text{interlock}})$ are fixed constants whose values are reported in Table 1.

Reward The per-step reward used to train the TQC policy is the weighted sum

$$r_t = \sum_j w_j \phi_j(t), \quad (31)$$

where the feature set and weights are detailed in Table 1. Key features include: an overtake success detector $\phi_{\text{ot}} = \mathbb{I}[\Delta s_{t-1} > \Delta s_{\text{ot}}^+ \wedge \Delta s_t < -\Delta s_{\text{ot}}^-]$ with hysteresis margins $(\Delta s_{\text{ot}}^+, \Delta s_{\text{ot}}^-)$; an approach incentive $\phi_{\text{app}} = \mathbb{I}[g < g_{\text{app}} \wedge \dot{g} < 0] / (1 + g)$ that encourages closing on a distant leader; and a geometric smoothness penalty $\phi_{\text{gs}} = (\Delta L_t^{\text{cap}})^2 + (\Delta R_t^{\text{cap}})^2 + (\Delta b_{\text{lat},t})^2$ that suppresses corridor jitter between consecutive steps.

The policy is optimized with Truncated Quantile Critics. Each critic represents a distributional action value by N_q quantile atoms $\{z_{\phi_i}^j(\tilde{\mathbf{o}}, \mathbf{u})\}_{j=1}^{N_q}$; the truncated target distribution discards the largest atoms before Bellman backup,

$$\mathcal{T}_{\text{trunc}} = \text{sort}\left(\{z_{\phi_i}^j(\tilde{\mathbf{o}}', \mathbf{u}') - \alpha \log \pi_{\theta}(\mathbf{u}'|\tilde{\mathbf{o}}')\}_{i,j}\right)_{1:K}, \quad (32)$$

where $K < N_q$ is the number of retained quantile atoms. The critic minimizes the quantile Huber loss

$$J_Q(\phi) = \sum_{i,j} \mathbb{E}_{\mathcal{D}, y \in r + \gamma \mathcal{T}_{\text{trunc}}} \left[\rho_{\tilde{\tau}_j}^{\kappa} \left(y - z_{\phi_i}^j(\tilde{\mathbf{o}}, \mathbf{u}) \right) \right], \quad (33)$$

while the actor maximizes the mean truncated value with an entropy regularizer,

$$J_{\pi}(\theta) = \mathbb{E}_{\tilde{\mathbf{o}} \sim \mathcal{D}, \mathbf{u} \sim \pi_{\theta}} \left[\alpha \log \pi_{\theta}(\mathbf{u}|\tilde{\mathbf{o}}) - \frac{1}{K} \sum_{z \in \mathcal{T}_{\text{trunc}}(\tilde{\mathbf{o}}, \mathbf{u})} z \right]. \quad (34)$$

The temperature α is adjusted automatically to maintain a target entropy. Training hyperparameters are summarized in Table 1.

One training episode consists of a single overtake duel: an NPC opponent is spawned ahead of the ego, and the episode terminates on successful overtake, collision, off-track excursion, loss of V2V communication, or a timeout. All quantities appearing in the simulation analysis (Sec. 7) use the notation established in this section: $\tilde{\mathbf{o}}_t$ for the map-conditioned observation, $\boldsymbol{\rho}(s)$ for the track prior, $\mathbf{a}_t$ and q_{side} for the RL action, and m_t for the FSM mode that generates the corridor.

3.2 Heading-Aware Corridor Carving

The corridor carver is a deterministic function

$$\begin{aligned} \mathcal{F}_{\text{carve}} : (\mathbf{o}_t, \mathbf{a}_t, m_t, \{\mathbf{o}_t^{(j)}\}_{j=1}^{N_{\text{opp}}}) \\ \mapsto \{(s_k, L_{\min,k}, L_{\max,k})\}_{k=0}^{N_{\text{sta}}-1} \end{aligned} \quad (35)$$

that converts opponent states and the FSM mode into a corridor of N_{sta} stations with uniform spacing Δs_{sta} over the planning horizon d_p . All numerical constants referenced below are collected in Table 1.

Heading-aware projection For opponent j with Frenet heading error $\chi^{(j)}$ and rectangular footprint of length L_v and width W_v , the half-extents in the Frenet (s, n) axes are

$$\begin{aligned} \rho_n^{(j)} &= \frac{1}{2} (L_v |\sin \chi^{(j)}| + W_v |\cos \chi^{(j)}|), \\ \rho_s^{(j)} &= \frac{1}{2} (L_v |\cos \chi^{(j)}| + W_v |\sin \chi^{(j)}|). \end{aligned} \quad (36)$$

The lateral exclusion radius around opponent j is then

$$\rho_{\text{exc}}^{(j)} = \rho_n^{(j)} + \frac{W_v}{2} + d_{\text{safe}}, \quad (37)$$

where d_{safe} is the RL-modulated safety margin from $\mathbf{a}_t$.

Spatial fading and startup protection Opponent influence is attenuated by a smooth fading kernel over distance,

$$\phi_{\text{fade}}(\Delta s) = \begin{cases} \cos^{p_{\text{fade}}} \left(\frac{\pi}{2} \frac{|\Delta s|}{S_{\text{fade}}} \right), & |\Delta s| < S_{\text{fade}}, \\ 0, & \text{otherwise,} \end{cases} \quad (38)$$

with characteristic range S_{fade} . A linear startup ramp $\phi_{\text{sr}}(k)$ prevents corridor collapse in the first few stations immediately ahead of the ego, where the planner has insufficient spatial authority to react.

Mode-conditioned corridor update Let $L_{\max,k}^{(0)}, L_{\min,k}^{(0)}$ be the geometric track-edge half-widths at station k . In OVERTAKE mode with committed pass side $\sigma \in \{+1, -1\}$, the carver tightens the opponent-side bound,

$$L_{\sigma,k} \leftarrow \sigma \left(\sigma n_{o,k} + \rho_{\text{exc}}^{(j)} \phi_{\text{fade}} \phi_{\text{sr}} \phi_{\text{rmp}} \right), \quad (39)$$

where $n_{o,k}$ is the predicted opponent lateral position at s_k and $\phi_{\text{rmp}}(k)$ is a distance-dependent blending ramp. In SHADOW mode the corridor is constructed as an asymmetric funnel that widens from the ego toward the pass side; the funnel width at convergence and the near-ego bonus are controlled by the parameters $(W_{\text{fun}}, W_{\text{near}}, \lambda_{\text{side}})$, with λ_{side} governing the pass-side allocation fraction. Across all modes a minimum corridor width $L_{\min}^{\text{corr}}$ is enforced, and the resulting bound profile is temporally smoothed by an exponential moving average with coefficient $\beta \in (0, 1)$,

$$[L_{\min,k}, L_{\max,k}]_t \leftarrow \beta [\cdot]_t + (1 - \beta) [\cdot]_{t-1}. \quad (40)$$

RL bias blending The RL action also emits a terminal lateral bias b_{lat} and global caps $(L_t^{\text{cap}}, R_t^{\text{cap}})$. These are blended into the carver output through a quadratic weight w_k that vanishes at the ego ($k = 0$) and reaches unity at a short look-ahead index k_{blend} ,

$$w_k = (k/k_{\text{blend}})^2 \mathbb{K}[k < k_{\text{blend}}] + \mathbb{K}[k \geq k_{\text{blend}}], \quad (41)$$

$$L_{\max,k} \leftarrow (1 - w_k) L_{\max,k} + w_k \min(L_{\max,k}, b_{\text{lat}} + L_t^{\text{cap}}), \quad (42)$$

$$L_{\min,k} \leftarrow (1 - w_k) L_{\min,k} + w_k \max(L_{\min,k}, b_{\text{lat}} - R_t^{\text{cap}}). \quad (43)$$

This blending is the mechanism through which the RL action directly shapes the planner corridor; the safety margin and side decision enter upstream through the carver, and all three are ultimately realized as $\mathcal{C}_k$.

4 Local Planning Layer

Given the corridor $\mathcal{C} = \{(s_k, L_{\min,k}, L_{\max,k})\}$ from the tactical layer, the planner produces a curvature-smooth lateral profile $l(s)$ inside $\mathcal{C}$ via a SOCP, and a maximum-grip speed profile $v(s)$ via a forward-backward sweep on the GG super-ellipse. Both solvers are exact and convex on their respective subproblems; convexity is preserved by sequential linearization of the only nonlinearity in (17).

4.1 SOCP Lateral Path Planning

Decision variable At each of N stations $k = 0, \dots, N - 1$ along the look-ahead horizon d_p , with uniform spacing $\Delta s_k = d_p/(N - 1)$, the planner instantiates four scalars,

$$\mathbf{y}_k = [l_k, \alpha_k, \zeta_k, \sigma_k]^\top \in \mathbb{R}^4, \quad (44)$$

namely lateral offset l_k , heading deviation α_k , curvature slack ζ_k , and the second-order cone witness σ_k . The stacked decision vector is $\mathbf{y} = [\mathbf{y}_0^\top, \dots, \mathbf{y}_{N-1}^\top]^\top \in \mathbb{R}^{4N}$. For numerical conditioning, all quantities are non-dimensionalized by a characteristic length r_{sc} and the derived speed $v_{\text{sc}} = \sqrt{r_{\text{sc}} g}$.

Cost The cost penalizes only the cone witness,

$$J_{\text{SOCP}}(\mathbf{y}) = \sum_{k=0}^{N-1} w_\sigma \sigma_k, \quad (45)$$

which, together with the cone constraint defined below, implements an L_2 penalty on the curvature slack. Path smoothness is induced indirectly by the dynamic equality: $l_{k+1} - l_k$ is slaved to $\alpha_k \Delta s_k$, which is in turn coupled to σ_k through (17); minimizing σ_k therefore minimizes a quadratic surrogate of $l''(s)$.

Equality constraints Initial and (mode-dependent) terminal conditions pin the boundary,

$$l_0 = l_{\text{ego}}, \quad \alpha_0 = \alpha_{\text{ego}}, \quad l_{N-1} = l_f^*, \quad \alpha_{N-1} = \alpha_f^*, \quad (46)$$

where the terminal target is selected as

$$l_f^* = \begin{cases} l_{\text{raceline}}(s_{N-1}) & \text{race mode, in-corridor,} \\ \frac{1}{2}(L_{\max, N-1} + L_{\min, N-1}) & \text{tactical/out-corridor.} \end{cases} \quad (47)$$

The Frenet dynamics (17) are discretized by trapezoidal integration; for $i = 0, \dots, N-2$,

$$\left(\mathbf{I} + \frac{\Delta s_i}{2} \mathbf{A}_{1,i}\right) \boldsymbol{\xi}_i + \frac{\Delta s_i}{2} \mathbf{B}_{1,i} \boldsymbol{\eta}_i - \left(\mathbf{I} - \frac{\Delta s_i}{2} \mathbf{A}_{2,i}\right) \boldsymbol{\xi}_{i+1} + \frac{\Delta s_i}{2} \mathbf{B}_{2,i} \boldsymbol{\eta}_{i+1} = -\frac{\Delta s_i}{2} (\mathbf{b}_{1,i} + \mathbf{b}_{2,i}), \quad (48)$$

with $\boldsymbol{\xi}_i = [l_i, \alpha_i]^\top$, $\boldsymbol{\eta}_i = [\zeta_i, \sigma_i]^\top$ and $(\mathbf{A}_{j,i}, \mathbf{B}_{j,i}, \mathbf{b}_{j,i})$ the linearization (17) at endpoints $j \in \{1, 2\}$ of segment i .

Linear inequality constraints (corridor) For each station k , the corridor injected by the tactical layer is mapped into two affine half-planes,

$$L_{\min,k} + d_{\text{veh}} \leq l_k \leq L_{\max,k} - d_{\text{veh}}, \quad (49)$$

where d_{veh} accounts for the half-vehicle width plus a static safety margin. In race mode without external corridor injection, $L_{\min,k}, L_{\max,k}$ default to the geometric track edges reduced by the on-line obstacle clearance.

Second-order cone constraint The curvature slack and its epigraph variable are bound through a single 3-dimensional cone per station,

$$\left\| \begin{bmatrix} 2\zeta_k, \sigma_k - 1 \end{bmatrix}^\top \right\|_2 \leq \sigma_k + 1, \quad k = 0, \dots, N-1, \quad (50)$$

which is equivalent to $\zeta_k^2 \leq \sigma_k$ for $\sigma_k \geq 0$. Combined with the linear cost in (45), the constraint implements a convex quadratic epigraph penalty on curvature slack without introducing a dense Hessian in the conic program.

Compiled SOCP Defining $\mathbf{c} \in \mathbb{R}^{4N}$ with $\mathbf{c}_{4k+3} = w_\sigma$ and zero elsewhere, $\mathbf{G} \in \mathbb{R}^{(2N+3N) \times 4N}$ stacking the corridor and cone rows, and $\mathbf{A} \in \mathbb{R}^{(2+2(N-1)+2) \times 4N}$ stacking (46)–(48), the planner solves

$$\begin{aligned} \mathbf{y}^* &= \arg \min_{\mathbf{y} \in \mathbb{R}^{4N}} \mathbf{c}^\top \mathbf{y} \\ \text{s.t. } \mathbf{A}\mathbf{y} &= \mathbf{b}, \\ \mathbf{G}\mathbf{y} &\preceq_{\mathcal{K}} \mathbf{h}, \end{aligned} \quad (51)$$

where $\mathcal{K} = \mathbb{R}_+^{2N} \times \mathcal{Q}_3^N$ is the product of the linear inequality cone and N second-order cones of dimension three. Both $\mathbf{G}$ and $\mathbf{A}$ are assembled directly in compressed column storage.

Sequential linearization The matrices in (17) depend on the operating point $(l_i, \alpha_i, \kappa_i)$. We resolve this by sequential linearization: at iteration j , (51) is solved with $(l_i^{(j)}, \alpha_i^{(j)})$ as the linearization point; the new iterate $(l_i^{(j+1)}, \alpha_i^{(j+1)})$ is read from $\mathbf{y}^*$; convergence is declared when

$$\max_i |l_i^{(j+1)} - l_i^{(j)}| < \varepsilon_l, \quad \max_i |\alpha_i^{(j+1)} - \alpha_i^{(j)}| < \varepsilon_\alpha, \quad (52)$$

or the iteration cap $J_{\max}$ is reached. In practice, two to three iterations suffice once the iterate is warm-started from the previous planning cycle's solution.

4.2 Forward-Backward GG-Coupled Velocity Profiling

The path $l^*(s)$ obtained from (51) is converted into a 2D curve $\mathbf{p}(s) = \mathbf{r}(s) + l^*(s)\mathbf{n}(s)$ whose curvature $\kappa^*(s)$ is recomputed analytically. The FB sweep then assigns a speed profile $v(s)$ that maximizes lap pace while respecting the super-elliptic GG envelope (18) and the speed-dependent grip caps $a_y^{\max}(v)$, $a_x^{\max}(v)$.

Curvature ceiling At each station the maximum stationary speed is bounded by lateral grip, $v_{\text{ce}}^2(s) = a_y^{\max}(v_{\text{ce}})/|\kappa^*(s)|$. The implicit dependence of $a_y^{\max}$ on v is resolved by fixed-point iteration,

$$v_{\text{ce}}^{(j+1)} = \sqrt{a_y^{\max}(v_{\text{ce}}^{(j)})/|\kappa^*(s)|}, \quad (53)$$

with stopping tolerance ε_v , typically converging in fewer than five iterations. The result is the station-wise curvature-ceiling array $V_{\text{ce}}[k]$.

Forward pass (acceleration) Initialized at $V_f[0] = \min(v_{\text{ego}}, V_{\text{ce}}[0])$, the forward sweep integrates from $k = 0$ to $N - 2$ as

$$a_{y,k} = \kappa_k^* V_f[k]^2, \quad (54)$$

$$a_{x,k}^{\text{avail}} = a_x^{\text{max}}(V_f[k]) \times [1 - (|a_{y,k}|/a_y^{\text{max}}(V_f[k]))^p]^{1/p}, \quad (55)$$

$$a_{x,k}^{\text{net}} = \min(a_{x,k}^{\text{avail}}, a_{x,k}^{\text{engine}}) + a_{\text{drag}}(V_f[k]), \quad (56)$$

$$V_f[k+1] = \min(\sqrt{V_f[k]^2 + 2\Delta s_k a_{x,k}^{\text{net}}}, V_{\text{ce}}[k+1]), \quad (57)$$

where $a_{x,k}^{\text{engine}}$ is the powertrain ceiling read from the PTP-on/off lookup tables and $a_{\text{drag}}(\cdot)$ is the drag model of (20). This sweep encodes the maximum-grip acceleration policy under combined slip.

Backward pass (braking) Terminated at $V_b[N-1] = \min(v_f, V_{\text{ce}}[N-1])$, the backward sweep runs from $k = N-2$ down to 0,

$$\tilde{a}_{y,k+1} = \kappa_{k+1}^* V_b[k+1]^2, \quad (58)$$

$$\tilde{a}_{x,k+1}^{\text{brk}} = a_x^{\text{max}}(V_b[k+1]) \times [1 - (|\tilde{a}_{y,k+1}|/a_y^{\text{max}}(V_b[k+1]))^p]^{1/p}, \quad (59)$$

$$\tilde{a}_{x,k+1}^{\text{tot}} = \tilde{a}_{x,k+1}^{\text{brk}} - a_{\text{drag}}(V_b[k+1]), \quad (60)$$

$$V_b[k] = \min(\sqrt{V_b[k+1]^2 + 2\Delta s_k \tilde{a}_{x,k+1}^{\text{tot}}}, V_{\text{ce}}[k]). \quad (61)$$

Drag enters with opposite sign to (56) because it *assists* braking; this sign discipline is essential to match on-track coastdown traces above 50 m/s.

Profile fusion and acceleration extraction The final speed profile is the pointwise minimum,

$$V[k] = \min\{V_f[k], V_b[k], V_{\text{ce}}[k]\}, \quad (62)$$

clipped to $[v_{\min}, v_{\max}]$ for numerical stability. Tangential and normal accelerations follow from energy conservation,

$$a_{T,k} = \frac{V[k+1]^2 - V[k]^2}{2\Delta s_k}, \quad a_{N,k} = \kappa_k^* V[k]^2. \quad (63)$$

A runtime aggressiveness factor $\eta \in (0, 1]$ and an upstream velocity cap v_{rc} are applied as the final scaling and clipping,

$$V_{\text{out}}[k] = \min(\eta V[k], v_{\text{rc}}). \quad (64)$$

The output trajectory $\{(x_k, y_k, \psi_k, \kappa_k, V_{\text{out}}[k], t_k)\}$, with time stamps $t_{k+1} = t_k + \Delta s_k / V_{\text{out}}[k]$, is resampled at the controller cadence Δt_{out} to produce a reference of N_{out} points.

5 Control Layer: MPCC

The controller closes the loop on the planner output with a Model Predictive Contouring Controller (MPCC) that exploits the dynamic model (10) and explicitly trades tracking accuracy against arc-length progress through a single positively-weighted terminal term.

5.1 Stage Cost and Progress Maximization

At each control cycle of period $T_c = 0.05$ s, the controller solves a finite-horizon nonlinear program over $N_c = 15$ stages, with a horizon of $T_c N_c = 0.75$ s. The stage cost couples a weighted lateral, yaw and speed tracking error with input regularization,

$$L_k(\mathbf{x}_k, \mathbf{u}_k, \zeta_k) = q_l e_{l,k}^2 + q_\psi e_{\psi,k}^2 + q_v e_{v,k}^2 + R_a a_k^2 + R_\delta \dot{\delta}_k^2 + w_\zeta \zeta_k^2, \quad (65)$$

where the contouring/lag decomposition is

$$e_{l,k} = -\sin \psi_{r,k}(p_{x,k} - x_{r,k}) + \cos \psi_{r,k}(p_{y,k} - y_{r,k}), \quad (66)$$

$$e_{\psi,k} = \psi_k - \psi_{r,k}, \quad e_{v,k} = v_{x,k} - v_{r,k}, \quad (67)$$

with $(x_{r,k}, y_{r,k}, \psi_{r,k}, v_{r,k})$ the reference quadruple from the planner expressed in the body frame at $k = 0$ to avoid the ill-conditioning of large global positions. The reference speed entering $e_{v,k}$ is clamped per stage,

$$v_{r,k} \leftarrow \text{clamp}(v_{r,k}, v_x - \Delta v_{\text{lo}}, v_x + \Delta v_{\text{hi}}), \quad (68)$$

which prevents the velocity-tracking term from dominating during sharp deceleration phases where the planner naturally lags behind the actual vehicle speed. The asymmetric bounds $(\Delta v_{\text{lo}}, \Delta v_{\text{hi}})$ are chosen such that the controller is permitted to overshoot the reference but penalized for undershooting.

The terminal cost is the unique asymmetric term,

$$\Phi(\mathbf{x}_{N_c}) = -q_{\text{prog}} s_{N_c}, \quad q_{\text{prog}} = 0.5, \quad (69)$$

which rewards arc-length progress along the prediction window. Because s is integrated as $\dot{s} = v_x$ in (1), maximizing s_{N_c} is equivalent to maximizing the path-tangent component of average speed over the horizon, weighted toward the end of the horizon. Combined with the friction-circle constraint introduced below, this term is the mechanism by which the controller produces the canonical entry-deceleration / apex-grip / exit-acceleration pattern without an explicit corner-by-corner reference shaping.

The MPCC weights are summarized in Table 1; they were tuned once on the global raceline and held constant across all races.

5.2 Friction-Circle Constrained NLP and Real-Time Solution

Constraints Equality constraints enforce the dynamics and the initial state,

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N_c - 1; \quad \mathbf{x}_0 = \mathbf{x}_{\text{init}}. \quad (70)$$

Box constraints on inputs and states are applied at every stage,

$$\underline{a} \leq a_k \leq \bar{a}, \quad -\dot{\delta}_{\text{max}} \leq \dot{\delta}_k \leq \dot{\delta}_{\text{max}}, \quad (71)$$

$$0 \leq v_{x,k} \leq v_{x,\text{max}}, \quad -\delta_{\text{max}} \leq \delta_k \leq \delta_{\text{max}}. \quad (72)$$

Combined slip on the rear axle is enforced as a single soft second-order cone per stage,

$$(m a_k)^2 + (C_r \alpha_{r,k})^2 \leq (\mu F_{z,r} + \zeta_k)^2, \quad \zeta_k \geq 0, \quad (73)$$

with the rear normal load determined by static weight distribution,

$$F_{z,r} = \frac{mg l_f}{l_f + l_r}. \quad (74)$$

The slack ζ_k in (73) is the same variable that appears in L_k through $w_\zeta \zeta_k^2$; this places a quadratic price on exceeding the friction envelope and is the primary mechanism by which the controller stays inside (18) without rendering the NLP infeasible during transient over-grip events.

Compiled NLP Collecting the decision variables $(\mathbf{X}, \mathbf{U}, \boldsymbol{\zeta}) \in \mathbb{R}^{8(N_c+1)} \times \mathbb{R}^{2N_c} \times \mathbb{R}_+^{N_c}$, the controller solves

$$\begin{aligned} \min_{\mathbf{X}, \mathbf{U}, \boldsymbol{\zeta}} \quad & \sum_{k=0}^{N_c-1} L_k(\mathbf{x}_k, \mathbf{u}_k, \zeta_k) + \Phi(\mathbf{x}_{N_c}) \\ \text{s.t.} \quad & (70)-(73) \text{ hold.} \end{aligned} \quad (75)$$

The NLP is solved by a primal-dual interior-point method with adaptive barrier strategy, warm-started from the previous cycle's solution shifted by one stage. An iteration limit and a wall-time budget are imposed to guarantee deterministic returns within the control period. The steering-angle and acceleration commands are reconstructed from the optimal solution as

$$\delta_0^* = \delta_{\text{prev}} + T_c \dot{\delta}_0^*, \quad a_0^* = a^*. \quad (76)$$

Real-time monitoring and fallback The MPCC solution is accepted only when (i) the solver returns a success status, (ii) the commanded acceleration and steering angle lie within predefined safety bounds $|a_0^*| \leq a_{\text{max}}$ and $|\delta_0^*| \leq \delta_{\text{max}}$, and (iii) the body speed is above the minimum threshold for which the linearized tire model in (5) remains valid. If any check fails, the controller reverts to a legacy lateral MPC plus longitudinal PI cascade for the current cycle and reattempts MPCC on the next cycle.

6 Real-Time Integration

The hierarchy is implemented as an acyclic cascade with explicit data contracts between adjacent layers. At tactical time t , the policy and FSM publish a guidance tuple

$$\begin{aligned}\mathcal{G}_t &= (m_t, V_{\text{cap},t}, \mathcal{C}_t, \pi_t), \\ \mathcal{C}_t &= \{(s_k, L_{\min,k}, L_{\max,k})\}_{k=0}^{N_{\text{sta}}-1},\end{aligned}\tag{77}$$

where m_t selects the terminal condition in (47), $V_{\text{cap},t}$ clips the FB speed profile through (64), $\mathcal{C}_t$ defines the lateral half-plane constraints (49), and π_t stores race-control permissions such as point-to-point passing availability. The planner holds the last valid guidance tuple if a tactical update is delayed; thus the feasible set evolves piecewise smoothly rather than discontinuously.

The planner converts $\mathcal{G}_t$ into a time-parameterized reference

$$\mathcal{R}_t = \{(x_k, y_k, \psi_k, \kappa_k, V_k, \tau_k)\}_{k=0}^{N_{\text{out}}-1},\tag{78}$$

after the SOCP path solve and GG-constrained velocity sweep. The MPCC controller transforms $\mathcal{R}_t$ into the local body frame at each control cycle, interpolates the reference on its prediction grid, and solves (75). This design leaves only two forms of authority flowing downstream: the tactical layer may restrict the feasible set and speed cap, and the planner may update the reference trajectory. Neither layer can overwrite controller constraints, tire limits, or actuator bounds.

The temporal hierarchy can therefore be summarized as

$$\underbrace{\text{strategy}}_{\text{tactical FSM tenure}} \succ \underbrace{\text{corridor}}_{\text{tactical/planner I/O}} \succ \underbrace{\text{MPCC}}_{\text{control loop}},\tag{79}$$

with each upper layer setting the admissible set of the layer below. This is the principal systems property used in the experimental analysis: tactical success is judged by whether the learned corridor is executable by the unchanged planner and controller, not by whether a learned controller happens to survive a particular duel.

7 Simulation Results

This section evaluates the proposed tactical–planning–control hierarchy in the semi-physical simulation environment used by the Abu Dhabi Autonomous Racing League (A2RL). The evaluation is organized as follows. First, the hardware-in-the-loop simulation environment and the communication interface are described. Second, all numerical parameters used by the vehicle model, tactical layer, planner, learning module, and controller are consolidated in a single table. Two representative overtaking cases are then analyzed in detail, followed by an aggregate 1000-lap validation.

7.1 Semi-Physical A2RL Simulation Environment

The closed-loop experiments were carried out in the official simulator developed by the A2RL race organizer. The simulator provides the Yas Marina track geometry, the Dallara EAV24 vehicle interface, the opponent vehicle, and the race-control message stream. The planner and controller were not coupled to the simulator through direct function calls. Instead, the software stack communicated with the virtual vehicle platform through a virtual CAN channel, using the same message abstraction as the race vehicle interface. Sensor states, opponent states, race-control flags, and vehicle feedback were received from the simulator, while planned actuation commands were returned through the same virtual CAN loop.

This configuration is a semi-physical simulation rather than a purely algorithmic benchmark: the decision stack runs as an external real-time process, all inter-layer and vehicle-interface timing must survive the message transport layer, and the controller receives only the state variables exposed by the platform interface. Consequently, failures caused by delayed tactical updates, infeasible planner outputs, or controller fallback logic appear in the closed loop exactly as they would in the vehicle software stack. The experiments therefore test not only whether an overtake trajectory exists, but whether the learned corridor, the SOCP planner, the speed profiler, and the MPCC controller remain mutually executable under the simulator’s virtual-CAN timing and race-control constraints.

7.2 Parameter Configuration

All numerical parameters used in the experiments are summarized in Table 1. The table intentionally reports only symbols and values so that the implementation settings can be inspected without interrupting the method derivations in the preceding sections.

Table 1: Parameter setting summary.

Sym.	Val.	Sym.	Val.	Sym.	Val.	Sym.	Val.	Sym.	Val.
m	742 kg	l_f, l_r	1.75, 1.35 m	I_z	1500 kg m ²	C_f, C_r	2.30, 2.40×10^5 N/rad	μ	1.20
$\delta_{\max}$	0.342 rad	$\dot{\delta}_{\max}$	0.5 rad/s	$[a, \bar{a}]$	$[-5, 8]$ m/s ²	$v_{x,\max}$	90 m/s	ε_v	0.5 m/s
$c_{d,2}$	$-1.04765 \text{ N s}^2/\text{m}^2$	$c_{d,1}$	47.5 N s/m	$c_{d,0}$	-1609.73 N	p	0.75	T_s	0.05 s
V_{char}	80 m/s	n_{char}	8 m	a_{char}	20 m/s ²	Δs_{char}	50 m	$\Delta V_{s,\text{char}}$	30 m/s
g_{char}	50 m	$\tau_{\text{tc},\max}$	30 s	ε_τ	0.01 m/s	L_t^{cap}	[1.5, 6.0]	R_t^{cap}	[1.5, 6.0]
d_{safe}	[0.2, 0.8]	V_{cap}	[15, 90]	g_{chase}	[18, 24]	$g_{\text{ot}}^{\text{front}}$	[16, 24]	$g_{\text{abort}}^{\text{rear}}$	[28, 45]
b_{lat}	$[-1.5, 1.5]$	w_{safe}	[1, 5]	q_{side}	$[-1, 1]$	$g_{\text{def}}^{\text{rear}}$	25 m	$g_{\text{def}}^{\text{rear}}$	20 m
Δn_{safe}	2.5 m	$g_{\text{interlock}}$	6 m	T_{imm}	30 s	ϕ_{ot}	30	ϕ_{ovb}	-20
ϕ_{col}	-100	ϕ_{lat}	-2	ϕ_{crush}	-5	ϕ_{app}	3	ϕ_{sm}	-0.1
ϕ_{gs}	-0.5	ϕ_{sbs}	0.5	ϕ_{wide}	0.25	α_Q	3×10^{-4}	$ \mathcal{D} $	10^5
B	256	γ	0.99	τ	5×10^{-3}	f_{feat}	51 $\rightarrow$ 256 $\rightarrow$ 256	$f_{\pi,Q}$	[256, 256]
N_q	25	K	20	d_p	300 m	N_{sta}	150	Δs_{sta}	2 m
p_{fade}	3	S_{fade}	50 m	W_{fun}	2.5 m	W_{near}	5 m	λ_{side}	0.6
$L_{\text{corr}}^{\text{min}}$	3 m	β	0.2	k_{blend}	9	d_{rmp}	40 m	k_{sr}	15
q_i	50	q_ψ	5	q_v	1	R_a	0.05	R_δ	30
q_{prog}	0.5	w_ζ	1000	N_c	15	T_c	0.05 s	$T_c N_c$	0.75 s

7.3 Scenario I: Short Fast-Closure Overtake

The first scenario is a short fast-closure pass. The ego vehicle enters the engagement with a large closing-speed advantage and must convert this advantage into a feasible lateral corridor before reaching the opponent. The run completes the pass with $\min \Delta s = -0.48 \text{ m}$, $\Delta t_{\text{pass}} = 1.13 \text{ s}$ after entering OVERTAKE, $\max V = 73.10 \text{ m/s}$, $\text{RMS}(e_l) = 0.02 \text{ m}$, $\text{RMS}(e_v) = 0.26 \text{ m/s}$, and tactical gain $\Delta \Gamma = 1.38$.

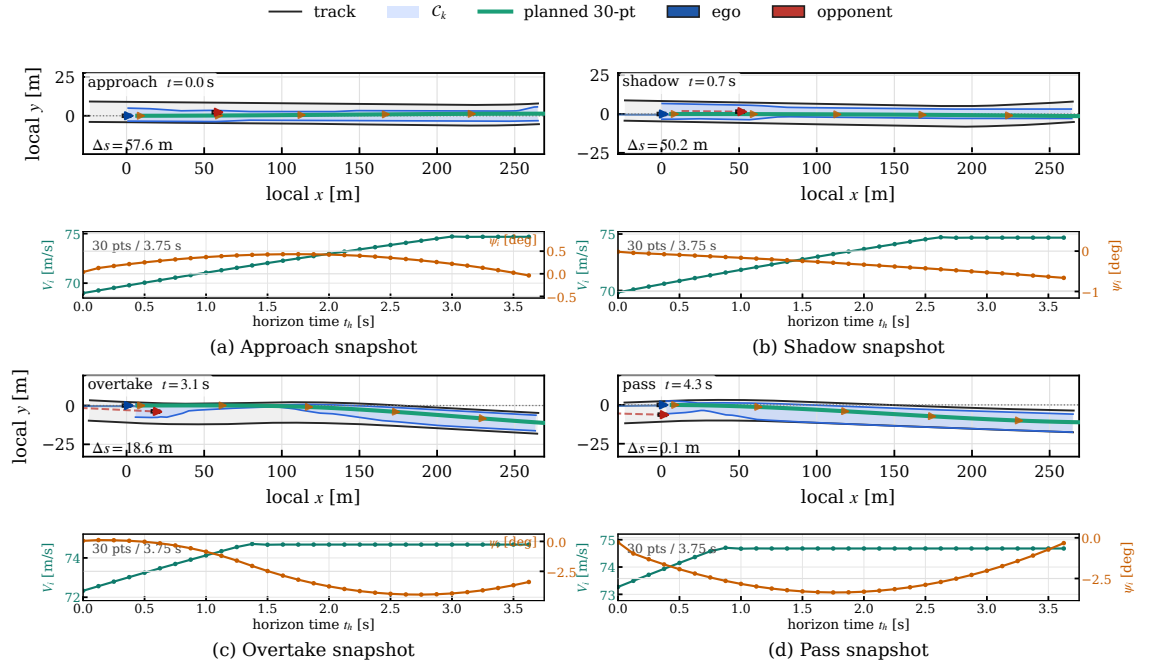


Figure 1: Scenario I engagement snapshots. The four representative instants show the planned tactical corridor and the local trajectory during approach, shadowing, commitment, and pass completion.

Figure 1 shows that the tactical decision is made before the vehicles become geometrically interlocked. In the first snapshot, the ego vehicle remains on a high-speed closure trajectory while the planner keeps the corridor centered enough to preserve recoverability. The second snapshot corresponds to the shadowing phase: the feasible set begins to open toward the selected pass side, but the corridor still contains a conservative fallback around the raceline. In the third snapshot, the FSM has committed to OVERTAKE; the corridor carver excludes the opponent footprint and forces the SOCP path to occupy the free side of the track. The final snapshot shows the pass completion, where the planned path has cleared the opponent without a discontinuous lateral jump. This progression is important because the learned component never commands steering or acceleration directly; it only reshapes the feasible set that the deterministic planner and MPCC must execute.

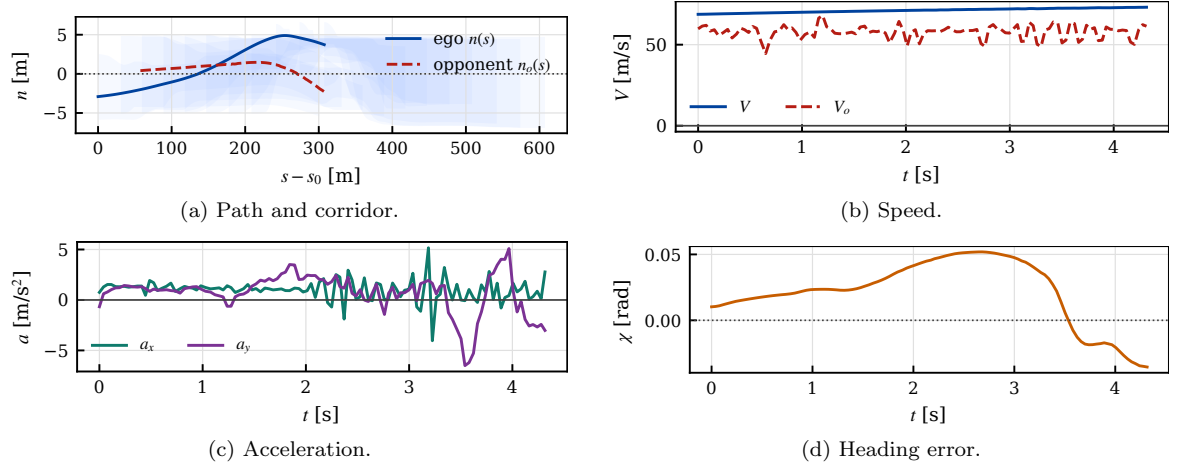


Figure 2: Scenario I closed-loop execution states.

Figure 2 quantifies the dynamic execution of the short pass. The path plot confirms that the ego trajectory remains inside the tactical corridor during the entire engagement, so the learned side decision does not create a geometric constraint violation. The speed trace reaches 73.10 m/s, indicating that the planner preserves the fast-closure advantage rather than sacrificing it through excessive caution. The longitudinal and lateral acceleration histories remain smooth during the transition into OVERTAKE, which shows that the GG-constrained speed profiler and MPCC friction slack prevent the maneuver from becoming a bang–bang lateral escape. The heading-error response stays bounded while the lateral path changes side, demonstrating that the corridor transition is converted into a controllable yaw demand rather than an abrupt heading correction.

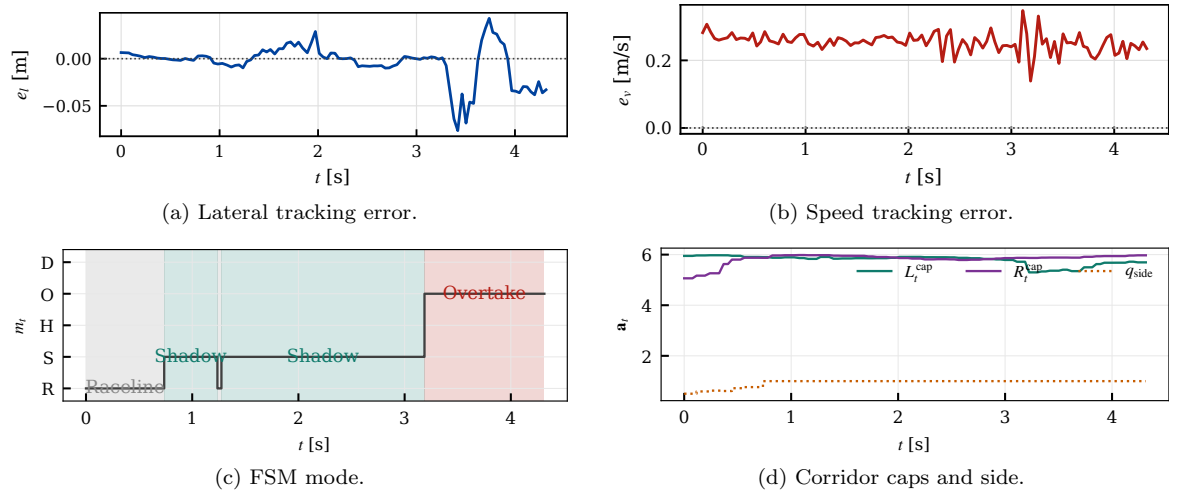


Figure 3: Scenario I tracking and tactical-state responses.

Figure 3 shows that the controller tracks the planner output with sub-decimeter accuracy despite the very short decision window. The lateral error has an RMS value of only 0.02 m, so the vehicle remains effectively on the planned path during the pass. The speed tracking error is also small, with $\text{RMS}(e_v) = 0.26$ m/s, which confirms that the velocity cap and the GG sweep are consistent with the MPCC's achievable acceleration envelope. The FSM mode plot shows a clean transition through SHADOW and OVERTAKE, while the corridor plot shows that the committed side is held during the pass rather than oscillating between alternatives. This stable side commitment is the key reason the planner can exploit the brief overtake opportunity without inducing high-frequency replanning.

7.4 Scenario II: Long Side-by-Side Overtake

The second scenario is a longer fast-closure pass in which the ego vehicle must remain side-by-side with the opponent for a longer interval before clearing it. The run completes the pass with

$\min \Delta s = -0.84$ m, $\Delta t_{\text{pass}} = 8.63$ s, $\max V = 65.83$ m/s, $\text{RMS}(e_l) = 0.05$ m, $\text{RMS}(e_v) = 1.82$ m/s, and tactical gain $\Delta \Gamma = 1.75$.

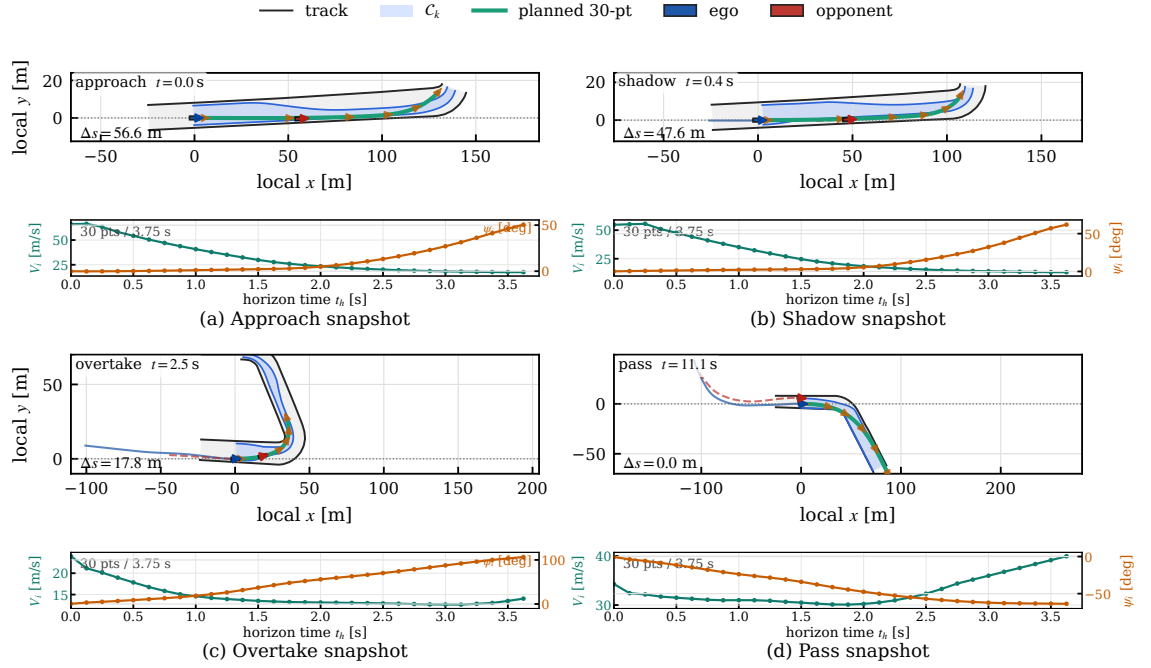


Figure 4: Scenario II engagement snapshots. The four representative instants show the planned tactical corridor and the local trajectory during the extended side-by-side overtake.

Figure 4 illustrates a qualitatively different engagement from Scenario I. The first instant shows the ego vehicle closing from behind while the corridor remains broad enough to retain both tactical options. The second instant captures the onset of lateral displacement: the planner starts to bias the path toward the selected side, but the speed profile remains conservative enough to avoid forcing an infeasible overlap with the opponent. During the third instant, the vehicles are side-by-side; the corridor excludes the opponent while preserving a continuous tube for the SOCP path. The fourth instant shows the ego vehicle ahead of the opponent, with the tactical layer relaxing the corridor after the pass. The sequence demonstrates that the same hierarchy handles not only short impulse-like passes, but also prolonged interactions where the corridor must remain stable for several seconds.

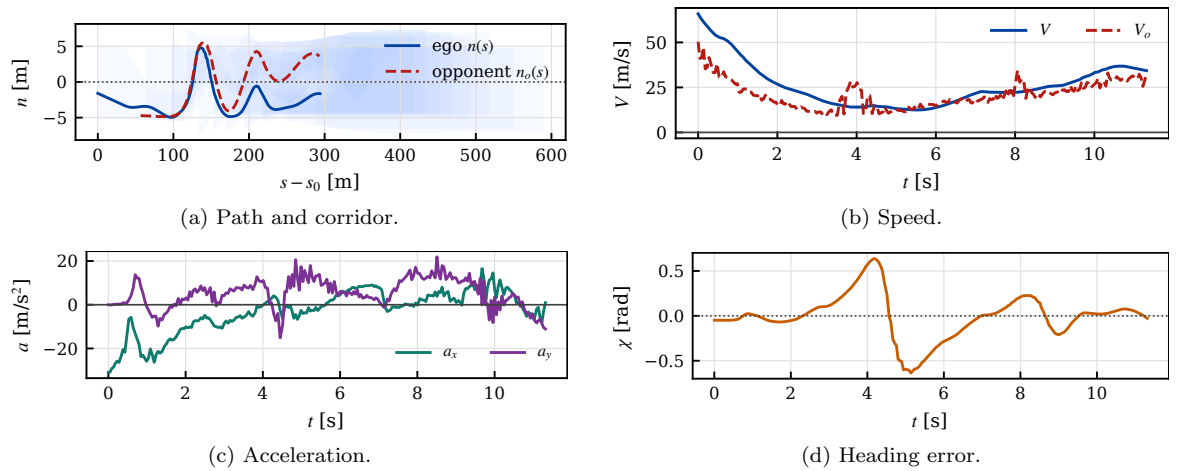


Figure 5: Scenario II closed-loop execution states.

Figure 5 shows that the longer engagement is executed with a lower peak speed of 65.83 m/s, reflecting the planner's need to remain in a viable side-by-side corridor rather than complete an

immediate speed-dominant pass. The path remains inside the carved corridor, which confirms that the tactical layer maintains a feasible spatial tube even when the opponent occupies the critical region for a longer time. The acceleration plot shows larger dynamic transients than Scenario I, as expected for an extended interaction, but the response remains smooth enough for the MPCC to track without friction-limit divergence. The heading response remains bounded during the side-by-side phase, indicating that the SOCP path regularization prevents the lateral maneuver from degenerating into a sharp steering correction.

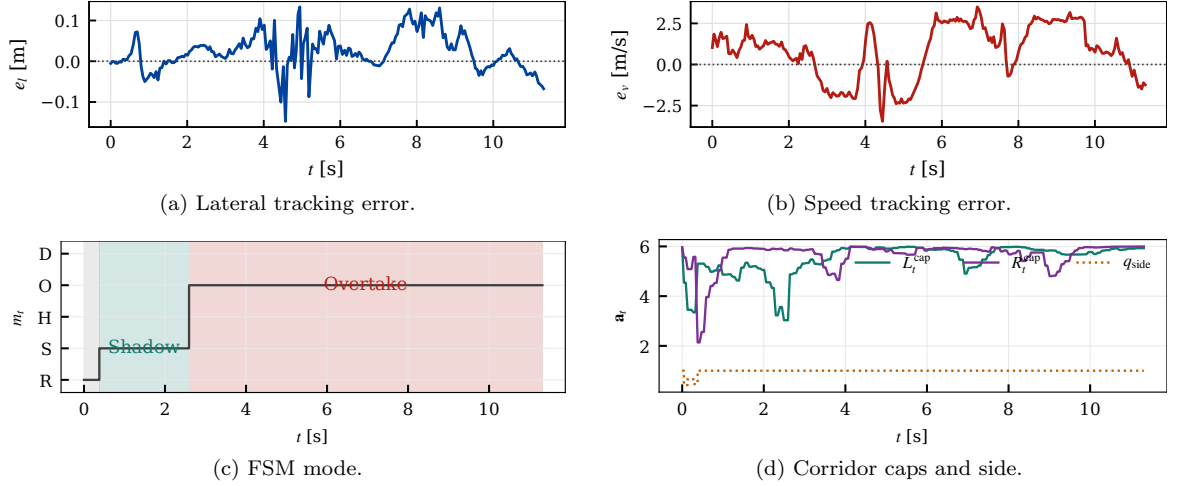


Figure 6: Scenario II tracking and tactical-state responses.

Figure 6 confirms that the increased duration of the duel does not destabilize the hierarchy. The lateral RMS tracking error is 0.05 m, still well below the corridor width and therefore small relative to the available passing margin. The speed error increases to 1.82 m/s because the controller must reconcile the speed profile with a longer side-by-side geometry, but the error remains bounded and does not prevent pass completion. The FSM plot shows a persistent OVERTAKE tenure rather than rapid mode switching, and the corridor plot shows a stable committed side throughout the decisive interval. This is the desired behavior for a learned tactical layer: once the policy selects a pass side, the downstream solvers receive a consistent feasible set long enough to execute the maneuver.

7.5 Simulation Summary

The two detailed scenarios expose complementary operating regimes. Scenario I tests whether the hierarchy can exploit a short closing-speed opportunity, whereas Scenario II tests whether the same architecture can maintain a stable side commitment during a longer side-by-side contest. Table 2 summarizes the closed-loop metrics of these two runs.

Table 2: Closed-loop metrics for the representative overtake scenarios.

Case	Δt_{pass} [s]	$\max V$ [m/s]	$\text{RMS}(e_l)$ [m]	$\text{RMS}(e_v)$ [m/s]	$\Delta\Gamma$ [-]
Scenario I	1.13	73.10	0.02	0.26	1.38
Scenario II	8.63	65.83	0.05	1.82	1.75

Beyond the two representative cases, the same frozen tactical policy, planner, and controller were evaluated over 1000 closed-loop laps in the A2RL simulator. No online learning, case-specific retuning, or controller weight adjustment was used during this validation. The aggregate results are reported in Table 3. The overtake success rate of 94.7% indicates that the learned tactical layer generalizes across a large number of repeated race interactions while preserving compatibility with the model-based planner and controller. Equally important, the validation was performed through the virtual-CAN semi-physical interface, so successful laps reflect end-to-end timing consistency rather than offline trajectory feasibility alone.

Overall, the experiments support the central design principle of the paper: learning is most effective when it determines the tactical feasible set, while the planner and controller enforce geometric consistency, handling-limit speed selection, and closed-loop vehicle execution. The short-pass and long-pass cases show that the architecture is not tuned to a single overtaking

Table 3: Aggregate semi-physical validation results.

Metric	Value
Closed-loop laps	1000
Successful overtake laps	947
Non-successful terminations	53
Overtake success rate	94.7%
Online learning during validation	No
Case-specific controller retuning	No

geometry, and the 1000-lap validation demonstrates that this separation remains robust under repeated semi-physical simulation.

8 Conclusion

This paper presented a hierarchical tactical-planning-control framework for wheel-to-wheel autonomous racing at Formula-class speeds. The central idea is to restrict learning to the tactical feasible set and to retain auditable model-based optimization for path planning, speed profiling, and closed-loop control. A map-conditioned TQC policy and a hysteretic FSM synthesize the overtake side, speed cap, and time-varying Frenet corridor; a heading-aware corridor carver converts the tactical decision into spatial constraints; a sparse SOCP computes a curvature-regularized path inside this corridor; a forward-backward sweep enforces the super-elliptic GG handling envelope; and an MPCC with terminal progress reward and soft rear-axle friction constraint executes the reference through the vehicle interface.

The resulting architecture separates tactical intent from physical execution. This separation is crucial in racing: the learned policy can choose where the vehicle is allowed to drive, but it cannot directly overwrite tire limits, actuator bounds, or controller dynamics. The semi-physical A2RL simulator experiments validate this design under a virtual-CAN communication loop rather than an offline trajectory benchmark. In the short fast-closure scenario, the system completed the pass 1.13 s after entering OVERTAKE, reached 73.10 m/s, and achieved a lateral RMS tracking error of 0.02 m. In the longer side-by-side scenario, the same frozen policy, planner, and controller completed the pass in 8.63 s with a lateral RMS error of 0.05 m. Across 1000 closed-loop semi-physical laps, the stack achieved a 94.7% overtake success rate without online learning or case-specific retuning.

These results support three conclusions. First, map-conditioned tactical learning can resolve the discrete side-selection and commitment problem that is difficult to encode in a single-vehicle raceline tracker. Second, constraining a convex local planner through a learned corridor provides a transparent interface between data-driven tactical reasoning and vehicle-dynamics feasibility. Third, the MPCC layer can execute distinct tactical corridors with sub-decimeter lateral accuracy when the upstream planner preserves curvature smoothness and handling-limit speed consistency. Future work will extend the validation to multi-opponent races, incorporate probabilistic opponent prediction tubes into the SOCP corridor, and transfer the semi-physical results to controlled track tests on the Dallara EAV24 platform.

References

- [1] F. Christ, A. Wischnewski, A. Heilmeier, and B. Lohmann, “Time-optimal trajectory planning for a race car considering variable tyre-road friction coefficients,” *Veh. Syst. Dyn.*, vol. 59, no. 4, pp. 588–612, 2021.
- [2] A. Heilmeier, A. Wischnewski, L. Hermansdorfer, J. Betz, M. Lienkamp, and B. Lohmann, “Minimum curvature trajectory planning and control for an autonomous race car,” *Veh. Syst. Dyn.*, vol. 58, no. 10, pp. 1497–1527, 2020.
- [3] N. R. Kapania, J. Subosits, and J. C. Gerdes, “A sequential two-step algorithm for fast generation of vehicle racing trajectories,” *ASME J. Dyn. Syst. Meas. Control*, vol. 138, no. 9, art. 091005, 2016.
- [4] T. Lipp and S. Boyd, “Minimum-time speed optimisation over a fixed path,” *Int. J. Control*, vol. 87, no. 6, pp. 1297–1311, 2014.
- [5] R. Verschuere, S. De Bruyne, M. Zanon, J. V. Frasch, and M. Diehl, “Towards time-optimal race car driving using nonlinear MPC in real-time,” in *Proc. IEEE Conf. Decision Control (CDC)*, 2014, pp. 2505–2510.

- [6] U. Rosolia and F. Borrelli, “Learning how to autonomously race a car: A predictive control approach,” *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 6, pp. 2713–2719, 2020.
- [7] M. Brunner, U. Rosolia, J. Gonzales, and F. Borrelli, “Repetitive learning model predictive control: An autonomous racing example,” in *Proc. IEEE Conf. Decision Control (CDC)*, 2017, pp. 2545–2550.
- [8] J. Kabzan, M. de la Iglesia Valls, V. Reijgwart *et al.*, “AMZ driverless: The full autonomous racing system,” *J. Field Robot.*, vol. 37, no. 7, pp. 1267–1294, 2020.
- [9] A. Wischniewski, M. Geisslinger, J. Betz *et al.*, “Indy autonomous challenge—Autonomous race cars at the handling limits,” in *Proc. FISITA World Automot. Congr.*, 2022, pp. 1–14.
- [10] J. Betz, T. Betz, F. Fent *et al.*, “TUM autonomous motorsport: An autonomous racing software for the Indy autonomous challenge,” *J. Field Robot.*, vol. 40, no. 4, pp. 783–809, 2023.
- [11] C. Jung, A. Finazzi, H. Seong *et al.*, “An autonomous system for head-to-head race: Design, implementation and analysis,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 10, pp. 10470–10486, 2023.
- [12] A. Raji, A. Liniger, A. Giove *et al.*, “A tricycle model to accurately control an autonomous racecar with locked differentials,” *IEEE Robot. Autom. Lett.*, vol. 9, no. 3, pp. 2499–2506, 2024.
- [13] J. Kabzan, L. Hewing, A. Liniger, and M. N. Zeilinger, “Learning-based model predictive control for autonomous racing,” *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 3363–3370, 2019.
- [14] L. Hewing, K. P. Wabersich, M. Menner, and M. N. Zeilinger, “Learning-based model predictive control: Toward safe learning in control,” *Annu. Rev. Control Robot. Auton. Syst.*, vol. 3, pp. 269–296, 2020.
- [15] A. Liniger, A. Domahidi, and M. Morari, “Optimization-based autonomous racing of 1:43 scale RC cars,” *Optim. Control Appl. Methods*, vol. 36, no. 5, pp. 628–647, 2015.
- [16] A. Liniger and J. Lygeros, “A noncooperative game approach to autonomous racing,” *IEEE Trans. Control Syst. Technol.*, vol. 28, no. 3, pp. 884–897, 2020.
- [17] A. Romero, S. Sun, P. Foehn, and D. Scaramuzza, “Model predictive contouring control for time-optimal quadrotor flight,” *IEEE Trans. Robot.*, vol. 38, no. 6, pp. 3340–3356, 2022.
- [18] T. Novi, A. Liniger, R. Capitani, and C. Annicchiarico, “Real-time control for at-limit handling driving on a predefined path,” *Veh. Syst. Dyn.*, vol. 58, no. 7, pp. 1007–1036, 2020.
- [19] R. Verschueren, G. Frison, D. Kouzoupis *et al.*, “acados—A modular open-source framework for fast embedded optimal control,” *Math. Program. Comput.*, vol. 14, pp. 147–183, 2022.
- [20] A. Domahidi and J. Jerez, “FORCES Professional,” Embotech AG, 2014–2024. [Online]. Available: <https://embotech.com/forces-pro>.
- [21] M. Wang, Z. Wang, J. Talbot, J. C. Gerdes, and M. Schwager, “Game-theoretic planning for self-driving cars in multivehicle competitive scenarios,” *IEEE Trans. Robot.*, vol. 37, no. 4, pp. 1313–1325, 2021.
- [22] W. Schwarting, T. Seyde, I. Gilitschenski, L. Liebenwein, R. Sander, S. Karaman, and D. Rus, “Deep latent competition: Learning to race using visual control policies in latent space,” in *Proc. Conf. Robot Learn. (CoRL)*, 2021, pp. 1855–1870.
- [23] D. Fridovich-Keil, E. Ratner, L. Peters, A. D. Dragan, and C. J. Tomlin, “Efficient iterative linear-quadratic approximations for nonlinear multi-player general-sum differential games,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2020, pp. 1475–1481.
- [24] F. Laine, D. Fridovich-Keil, C.-Y. Chiu, and C. J. Tomlin, “The computation of approximate generalized feedback Nash equilibria,” *SIAM J. Optim.*, vol. 33, no. 1, pp. 294–318, 2023.
- [25] R. Spica, E. Cristofalo, Z. Wang, E. Montijano, and M. Schwager, “A real-time game theoretic planner for autonomous two-player drone racing,” *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1389–1403, 2020.

- [26] A. Liniger and J. Lygeros, “Real-time control for autonomous racing based on viability theory,” *IEEE Trans. Control Syst. Technol.*, vol. 27, no. 2, pp. 464–478, 2019.
- [27] G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou, “Information-theoretic model predictive control: Theory and applications to autonomous driving,” *IEEE Trans. Robot.*, vol. 34, no. 6, pp. 1603–1622, 2018.
- [28] P. R. Wurman, S. Barrett, K. Kawamoto *et al.*, “Outracing champion Gran Turismo drivers with deep reinforcement learning,” *Nature*, vol. 602, no. 7896, pp. 223–228, 2022.
- [29] H. Jia, Y. Wu, X. Yang, J. Pan, and L. Liu, “Bayesian intent inference for head-to-head autonomous racing,” *IEEE Trans. Intell. Veh.*, vol. 8, no. 7, pp. 3851–3863, 2023.
- [30] F. Fuchs, Y. Song, E. Kaufmann, D. Scaramuzza, and P. Duerr, “Super-human performance in Gran Turismo Sport using deep reinforcement learning,” *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 4257–4264, 2021.
- [31] Y. Song, H. Lin, E. Kaufmann, P. Duerr, and D. Scaramuzza, “Autonomous overtaking in Gran Turismo Sport using curriculum reinforcement learning,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2021, pp. 9403–9409.
- [32] B. D. Evans, H. A. Engelbrecht, and H. W. Jordaan, “Safe reinforcement learning for high-speed autonomous racing,” *Cogn. Robot.*, vol. 3, pp. 107–126, 2023.
- [33] A. Brunnbauer, L. Berducci, A. Brandstätter *et al.*, “Latent imagination facilitates zero-shot transfer in autonomous racing,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2022, pp. 7513–7520.
- [34] S. He, J. Zeng, B. Zhang, and K. Sreenath, “Rule-based safety-critical control design using control barrier functions with application to autonomous lane change,” *IEEE Trans. Control Syst. Technol.*, vol. 31, no. 6, pp. 2701–2715, 2023.
- [35] Y. Ye, X. Zhang, and J. Sun, “Automated vehicle’s behavior decision making using deep reinforcement learning and high-fidelity simulation environment,” *Transp. Res. Part C*, vol. 107, pp. 155–170, 2019.
- [36] Z. N. Sunberg, C. J. Ho, and M. J. Kochenderfer, “The value of inferring the internal state of traffic participants for autonomous freeway driving,” in *Proc. Amer. Control Conf. (ACC)*, 2017, pp. 3004–3010.
- [37] B. Paden, M. Čáp, S. Z. Yong, D. Yershov, and E. Frazzoli, “A survey of motion planning and control techniques for self-driving urban vehicles,” *IEEE Trans. Intell. Veh.*, vol. 1, no. 1, pp. 33–55, 2016.
- [38] D. González, J. Pérez, V. Milanés, and F. Nashashibi, “A review of motion planning techniques for automated vehicles,” *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 4, pp. 1135–1145, 2016.
- [39] J. Betz, H. Zheng, A. Liniger *et al.*, “Autonomous vehicles on the edge: A survey on autonomous vehicle racing,” *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 458–488, 2022.
- [40] J. Betz, A. Wischniewski, A. Heilmeyer *et al.*, “A software architecture for the dynamic path planning of an autonomous racecar at the limits of handling,” in *Proc. IEEE Int. Conf. Connect. Veh. Expo (ICCVE)*, 2019, pp. 1–8.
- [41] E. Kaufmann, L. Bauersfeld, A. Loquercio *et al.*, “Champion-level drone racing using deep reinforcement learning,” *Nature*, vol. 620, no. 7976, pp. 982–987, 2023.
- [42] J. Ziegler, P. Bender, M. Schreiber *et al.*, “Making Bertha drive—An autonomous journey on a historic route,” *IEEE Intell. Transp. Syst. Mag.*, vol. 6, no. 2, pp. 8–20, 2014.
- [43] M. Werling, J. Ziegler, S. Kammel, and S. Thrun, “Optimal trajectory generation for dynamic street scenarios in a Frénet frame,” in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2010, pp. 987–993.
- [44] U. Rosolia and F. Borrelli, “Learning model predictive control for iterative tasks. A data-driven control framework,” *IEEE Trans. Autom. Control*, vol. 63, no. 7, pp. 1883–1896, 2018.

- [45] L. Brunke, M. Greeff, A. W. Hall *et al.*, “Safe learning in robotics: From learning-based control to safe reinforcement learning,” *Annu. Rev. Control Robot. Auton. Syst.*, vol. 5, pp. 411–444, 2022.
- [46] H. Lin, Y. Song, and D. Scaramuzza, “Reaching the limit in autonomous racing: Optimal control versus reinforcement learning,” *Sci. Robot.*, vol. 8, no. 82, art. eadg1462, 2023.
- [47] A. Remonda, E. Veas, and G. Luzhnica, “Comparing the performance of reinforcement learning algorithms for autonomous racing,” *Eng. Appl. Artif. Intell.*, vol. 118, art. 105692, 2023.